2021-0621-1111-3115
UNIVERSITY OF THE PHILIPPINES
3F, Quezon Hall, Diliman, Quezon City
Tele/Fax: (632) 8962-6345; 89818722; 89818500 local 2528
Email: ovpaa@up.edu.ph

Office of the Vice President for Academic Affairs

9 August 2021

HON. J. PROSPERO E. DE VERA III, DPA
Chairman
Commission on Higher Education
C.P. Garcia Ave., Quezon City

Subject: Endorsement of CHED-PCARI proposal of IHITM 2018-035: "Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)" headed by Dr. Michael Y. Roleda (UP Diliman) and Dr. William Gerwick (UC San Diego)

Dear Chairman De Vera,

May we respectfully inform you that the UP Board of Regents, in its 1362nd meeting last 29 July 2021, approved the CHED-PCARI project proposal titled, *"Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)"* to be implemented by the University of the Philippines Diliman (Dr. Michael Roleda) in collaboration with the University of California San Diego (Dr. William Gerwick).

The PhycoPRO project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweed species with high economic potential. This study will identify and prioritize species that have sufficient natural standing stock and species that are culturable or a cultivation technology can be developed and capitalize on their unique chemistry for commercial applications.

Following your instructions, we subjected the proposal to an expert reviewer who agrees with the CHED-PCARI technical reviewers that the proposal is 'good to go'! It is very well written, of profound significance to the livelihood of seaweed farmers and is backed by the impressive credentials, experience, and expertise of the Co-Project Investigators. AVPAA for Research Carla Dimalanta and her team also reviewed the line-item-budget (LIB) of the project and, except for two minor comments, which were relayed to the proponents found the LIB in order.

Attached herewith are the copies of the BOR certification, revised project proposal, LIB, and budget justification.

Thank you.

Truly yours,

Maria Cynthia Rose Banzon Bautista
Vice-President for Academic AffairsUniversity of the Philippines

Republic of the Philippines
Mindanao State University at Naawan
9023 Naawan, Misamis Oriental
http://www: msunaawan.edu.ph

Office of the Chancellor

March 28, 2022

PCARI Advisory Group and
J. PROSPERO E. DE VERA III, DPA
Chairperson
Commission on Higher Education
Higher Education Development Center (HEDC) Building
C. P. Garcia Avenue, Diliman, Quezon City

Dear Sir:

Greetings of peace and good will!

This letter is in reference to the proposed project of **Dr. Irene B. Rodriguez and Dr. Michael Y. Roleda** from UP MSI entitled, ***"Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)"*** for possible funding under the Commission on Higher Education Philippine-California Advanced Research Institutes (PCARI) Program. The overall goal of the project is to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts to coastal communities in particular, and mankind in general.

We would like to express our great enthusiasm for being chosen by the UP- Marine Science Institute (UP MSI) as partner university for this project. We believe that this will enhance our capability in implementing collaborative research endeavors.

After series of discussions, MSU at Naawan signifies our commitment and support for the implementation of the project. Likewise, we commit to assist the project team in delivering the target deliverables and participate in any research activities called by the project leader.

Thank you very much and more power!

Sincerely yours,

ELNOR C. ROA, Ph.D.
Chancellor II

Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

Office of the University President

March 28, 2022

HON. J. PROSPERO E. DE VERA III, DPA
Chairperson
Commission on Higher Education
Higher Education Development Center Building
C.P. Garcia Ave, Diliman, Quezon City

Through: HON. ALDRIN A. DARILAG, Ph.D., CHRP, RMT, RN
Commissioner, Commission on Higher Education

Sir:

This is to respectfully and formally express the commitment of Partido State University to participate as collaborating SUC in the CHED-PCARI program: **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO).** The proponents of this program are Dr. Michael Y. Roleda and **Dr. Irene B. Rodriguez** (Balik Scientist in ParSU) from the Marine Science Institute University of the Philippines; and, Dr. William H. Gerwick and Dr. Lena S. Gerwick from the Scripps Institution of Oceanography University of California San Diego California, USA.

As such, believing in the overall goal of this program to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts to coastal communities in particular, and mankind in general, **Partido State University will participate in all research activities and delivering outputs of the project.**

Thank you very much for this worthwhile opportunity.

Sincerely,

RAUL G. BRADECINA, Ph.D.
SUC President III

Republic of the Philippines
Sorsogon State University
Office of the President
Sorsogon City
Tel No. (056) 211-0200, email address: ssc@sorsogonstatecollege.edu.ph

23 February 2022

J. Prospero E. De Vera III, PhD
Chairman
Commission on Higher Education

RELEASED
02-24-22
3:54PM

Dear Chairman De Vera,

Greetings of peace!

This is to express our commitment as Collaborating Agency of the proposed project, **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)** lead by Dr Michael Y. Roleda and Irene B. Rodriguez of UP Diliman, The Marine Science Institute. Rest assured that we will exert full effort for its consummation.

Thank you very much.

Very truly yours,

HELEN R. LARA, Ed.D.
SUC President III
Sorsogon State University

Republic of the Philippines
UNIVERSITY OF EASTERN PHILIPPINES
University Town, Northern Samar, Philippines
Web: http://uep.edu.ph; Email: uepnsofficial@gmail.com

12 May 2022

J. PROSPERO E. DE VERA, III, DPA
Chairperson
Commission on Higher Education
Diliman 1101, Quezon City

Sir:

Greetings!

This is in connection to the proposed research project on **PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS (PhycoPRO)** to be implemented by the University of the Philippines - Marine Science Institute and University of California San Diego – Scripps Institution of Oceanography.

The University of Eastern Philippines through the Office of the University Board Secretary has submitted the said proposed research project to the Board of Regents for endorsement/clearance on the long-term commitment of this university to support basic and applied seaweed research.

This study will be beneficial to the university and its component communities as it will enhance the R&D capacity of UEP after the completion of the project, consider the feasibility of establishing a "seaweed nursery", and development of practical uses/applications and/or sustainability of the research beyond the duration.

We look forward to cooperating with your institution in the conduct of this research work. Thank you very much.

Very truly yours,

KARINA MILAGROS C. LIM, PhD
Vice President for
Research, Development,
and Extension
University of Eastern Philippines

UNIVERSITY OF THE PHILIPPINES

CERTIFICATION

This is to certify that during the **1362nd Meeting of the Board of Regents held on 29 July 2021**, the Board APPROVED the following:

> **"CHED-PCARI Project "Phytochemical Characterization of Macroalgae for Food and High Value Products (PhycoPRO)" by Dr. Michael Roleda and Dr. William Gerwick "**

WITNESS MY HAND AND SEAL this 5th day of August 2021 at Diliman, Quezon City.

ROBERTO M.J. LARA
Secretary of the University
and of the Board of Regents

Quezon Hall, University of the Philippines, Diliman, Quezon City 1101, Philippines
Tel. # (632) 927 67 80, Telefax # (632) 920 68 97, 920 68 85, e-mail osu@up.edu.ph

Republic of the Philippines
MINDANAO STATE UNIVERSITY
Marawi City

OFFICE OF THE SECRETARY OF THE UNIVERSITY
AND OF THE BOARD OF REGENTS

EXCERPTS FROM THE MINUTES OF THE 259th MEETING OF THE BOARD OF REGENTS HELD ON JUNE 13, 2022 VIA ZOOM TELECONFERENCING AND FACE-TO-FACE AT HOTEL KIMBERLY MANILA, 770 PEDRO GIL ST., MALATE, MANILA

x-x-x x-x-x x-x-x

"RESOLUTION NO. 239, S. 2022"

RESOLVED, that upon the recommendation of the President of the University and the Executive Committee of the Board, MSU at Naawan Naawan's project proposal entitled Phytochemical Characterization of Macroalgae to be submitted to the Commission on Higher Education, is hereby **APPROVED**.

APPROVED"

x-x-x x-x-x x-x-x

CERTIFIED TRUE AND CORRECT:

ATTY. SHIDIK T. ABANTAS
Secretary of the University
and of the Board of Regents

STA:rlf
24 June 2022
(Not valid without seal)

Republic of the Philippines
PARTIDO STATE UNIVERSITY
Goa, Camarines Sur

RESOLUTION PASSED DURING THE 84th REGULAR BOARD OF REGENTS MEETING HELD ON APRIL 4, 2022 AT CATANDUANES STATE UNIVERSITY

Resolution No. 34, s. 2022

WHEREAS, the Partido State University is mandated to undertake advanced studies, research, production and extension services, and provide leadership in these areas pursuant to Section 2 of RA 9029;

WHEREAS, to adhere to the mandate, Partido State University is currently hosting a Balik Scientist with specialization in marine biogeochemistry, metallomics, environmental biotechnology, and environmental analytical chemistry;

WHEREAS, the Balik Scientist is a Co-Project Leader in the project entitled, "Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)", which was submitted to CHED-PCARI for possible funding;

WHEREAS, the PhycoPRO project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal;

WHEREAS, the Partido State University have faculty researchers working on seagrass ecosystems and are interested in pursuing seaweeds research to facilitate introduction of seaweed farming to coastal communities in Camarines Sur;

WHEREAS, the Partido State University is willing to commit on this project and to sustain the quest for knowledge, even after the duration of the CHED-PCARI funding;

NOW THEREFORE, BE IT RESOLVED AS IT HEREBY RESOLVED TO APPROVE THE COMMITMENT OF PARTIDO STATE UNIVERSITY TO PARTICIPATE AS COLLABORATING SUC IN THE CHED-PCARI PROJECT: PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS (PHYCOPRO), SUBJECT TO THE FINAL APPROVAL OF THE COMMISSION ON HIGHER EDUCATION.

ADOPTED this 4th day of April, 2022 during the 84th Regular Board of Regents Meeting at Catanduanes State University, Virac, Catanduanes.

I hereby certify to the correctness of the afore-going approved resolution:

LEIH ANNE R. ODOÑO
Board Secretary V

ATTESTED:

RAUL G. BRADECINA, Ph.D.
SUC President III

cRepublic of the Philippines
SORSOGON STATE UNIVERSITY
Sorsogon City

May 24, 2022

THE HONORABLE MEMBERS
BOARD OF REGENTS
SORSOGON STATE UNIVERSITY

THRU **DR. ALDRIN A. DARILAG**
CHED Commissioner
Chairman, SorSU-BOR

SIR:

Greetings!

The Sorsogon State University in its commitment to realize its vision as a research university would like to pursue international research engagements which will significantly impact both in local and global scale. With this, SorSU is collaborating with the University of the Philippines, University of California San Diego and other State University and Colleges (SUCs) under the Philippine-California Advanced Research Institutes (PCARI) Project in the research titled Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO).

This research aims to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts on coastal communities and beyond. Its main objectives are: *(1) performing a phytochemical characterization of at least 50 seaweed species from the Philippines; (2) assess the potential anti-cancer activity of extracts and purified compounds from seaweed associated epiphytes;* and *(3) conduct training workshops and seminars on seaweed farming techniques, collection preservation methods, and propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders.* As a collaborating SUC, Sorsogon State University will assist in conducting the workshops and training for seaweed farmers, which will aid them in properly harvesting and preservation procedures to maximize financial earning capacity.

With this, the University hereby respectfully submit by way of this referendum **the request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.**

May this request merit your consideration. Thank you very much.

Very truly yours,

HELEN R. LARA, EdD
SUC President III

Comments:

Approved ☑
Disapproved ☐

ALDRIN A. DARILAG, PhD
CHED Commissioner
Chairman, SorSU-BOR

Republic of the Philippines
SORSOGON STATE UNIVERSITY
Sorsogon City

REFERENDUM NO. __ s. 2022
May 24, 2022

SUBJECT MATTER:

Request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

LEGAL BASES:

- Republic Act 10653 (Sorsogon State University Charter)
- Republic Act 8292 (Higher Education Modernization Act of 1997)
- Intellectual Property Code (RA No. 8293)
- Republic Act No. 9184 (Bidding Act)

SorSU BOR Members	ACTION	SIGNATURE
DR. ALDRIN A. DARILAG CHED Commissioner and Chair of the Board		
DR. HELEN R. LARA SorSU President and Vice Chair of the Board	Approved	
CONG. EVELINA G. ESCUDERO House of Representatives Committee on Higher and Technical Education	Approved	
HON. MA. BERNADETTE E. QUIRANTE Senate Committee on Education, Arts and Culture	Approved	
DIR. AGNES E. TOLENTINO MEOs BOR Director		
DIR. ROMMEL SERRANO LUCs BOR Director		
DIR. RODEL TORNILLA DA BOR Executive Director		
DR. JHONNER RICAFORT Alumni Regent		
HON. VINCE P. JANORAS Student Regent		
(Vacant) Faculty Regent		

FOR APPROVAL OF THE BOARD

Request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

CSW FORM

A. Proponent: Office of the President

B. Action Requested:

- For Approval of the Board

C. Rationale/ Background Information

The Sorsogon State University in its commitment to realize its vision as a research university would like to pursue international research engagements which will significantly impact both in local and global scale. With this, SorSU is collaborating with the University of the Philippines, University of California San Diego and other State University and Colleges (SUCs) under the Philippine-California Advanced Research Institutes (PCARI) Project in the research titled Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO).

This research aims to contribute to knowledge generation on the potential uses of seaweeds that will have societal and economic impacts on coastal communities and beyond. Its main objectives are: *(1) performing a phytochemical characterization of at least 50 seaweed species from the Philippines; (2) assess the potential anti-cancer activity of extracts and purified compounds from seaweed associated epiphytes;* and *(3) conduct training workshops and seminars on seaweed farming techniques, collection preservation methods, and propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders.* As a collaborating SUC, Sorsogon State University will assist in conducting the workshops and training for seaweed farmers, which will aid them in properly harvesting and preservation procedures to maximize financial earning capacity.

With this, the University hereby respectfully submit by way of this referendum the request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

D. Supporting Documents

- SorSU Letter of Commitment as Collaborating Agency
- PCARI Project Form No. 3b: Narrative Proposal
- Summary of the involvement of the collaborating SUC in the implementation of the activities under the PhycoPRO
- Highlights of the Minutes of the Meeting with CHED Comm. Darilag & Phycopro Proponents *(February 18, 2022)*

E. Legal Bases

- Republic Act 11088 (Sorsogon State University Charter)
- Republic Act 8292 (Higher Education Modernization Act of 1997)
- Intellectual Property Code (RA No. 8293)
- Republic Act No. 9147 (Wildlife Act)

F. Proposed BOR Resolution

Resolution No. __ s, 2022

Request for endorsement of the proposed long term commitment of SorSU in the collaborative initiative for the CHED-PCARI project: Phytochemical characterization of macroalgae for food and higher value products (PhycoPRO) for approval of the CHED Commission En Banc.

G. Certification

I hereby certify to the accurateness and completeness of all the attached documents and that all pertinent University and Government rules and regulations regarding the matter are fully complied with.

HELEN R. LARA, EdD
President

Republic of the Philippines
Sorsogon State University
Office of the President
Sorsogon City
Tel No. (056) 211-0260 email address: ssc@sorsogon.sti.edu.ph

25 February 2022

SORSOGON STATE UNIVERSITY
RELEASED
02-24-22

J. Prospero E. De Vera III, PhD
Chairman
Commission on Higher Education

Dear Chairman De Vera

Greetings of peace!

This is to express our commitment as Collaborating Agency of the proposed project, **Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)** lead by Dr Michael Y. Roleda and Irene B. Rodriguez of UP Diliman, The Marine Science Institute. Rest assured that we will exert full effort for its consummation.

Thank You very much.

Very truly yours,

HELEN R. LARA, Ed.D.
SUC President III
Sorsogon State University

Republic of the Philippines
UNIVERSITY OF EASTERN PHILIPPINES
University Town, Northern Samar
Web: http://uep.edu.ph Email: uepnsofficial@gmail.com

BOARD OF REGENTS
REFERENDUM NO. 04, S. 2022

APPROVAL OF THE REQUEST OF THE UNIVERSITY FOR FAVORABLE SUPPORT/ENDORSEMENT PERTAINING TO THE IMPLEMENTATION OF THE PHYTOCHEMICAL CHARACTERIZATION OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS WITH A PROPOSED Budget OF P 79, 959, 718.14

WHEREAS, the request for **favorable support/endorsement pertaining to the implementation of the Phytochemical Characterization of Macroalgae for Food and high value products with a proposed budget of p 79, 959, 718.14** has been presented to the Honorable Board of Regents;

WHEREAS, the university commits to support the government's goal towards the enhancement of the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint location where their growth will be maximal;

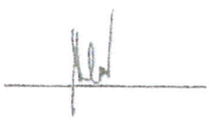
WHEREAS, that as a collaborating agency of the University of the Philippines-Diliman in the implementation of the Phytochemical Characterization Project of Macroalgae for Food and High Value Products amounting to **P 79, 959, 718.14** , the University of Eastern Philippines hereby expresses its favorable support to the said endeavor;

WHEREAS, that Honorable Board of Regents has reviewed the subject request and has found the same to be in order;

WHEREAS, the Administrative Committee endorsed the same for approval of the UEP Board of Regents through Finance Committee Referendum No. 01 Series of 2022;

NOW THEREFORE, BE IT RESOLVED AS IT IS HEREBY RESOLVED, THE APPROVAL VIA REFERENDUM OF THE UEP'S FAVORABLE SUPPORT TO THE IMPLEMENTATION OF THE PHYTOCHEMICAL CHARACTERIZATION PROJECT OF MACROALGAE FOR FOOD AND HIGH VALUE PRODUCTS AMOUNTING TO P 79, 959, 718.14.

DATE: **May 30, 2022**

	APPROVED	DISAPPROVED
HON. BONIFACIO G. UY Regional Director NEDA, Regional Office VIII		
HON. ERNESTO M. GRANADA Regional Director DOST, Regional Office VIII		
HON. SEARY ANNE C. BALLICUD President UEP Federation of Student Councils		
HON. ALBERT P. YRUMA President UEP Alumni Association Federation		
HON. ROMUALDO L. HIPE President UEP Federation of Faculty Unions		
HON. ANGEL C. ENRIQUEZ Regional Executive Director DA, Regional Field Office VIII		
HON. EMIL L. ONG (+) Representing: Hon. Mark O. GO Chairperson, Committee on Higher and Technical Education House of Representatives		

	APPROVED	DISAPPROVED
HON. DON L. ABALON Representing: Hon. Emmanuel Joel J. Villanueva Chairperson, Committee on Higher and Technical Education House of Representatives Senate of the Philippines	_____	_____
HON. CHERRY I. ULTRA President, UEP Vice Chair, Board of Regents	_____	_____
HON. ALDRIN A. DARILAG Commissioner, CHED Chairman, UEP Board of Regents	_____	_____

Attested:

DR. NELIA M. ADORA
Board Secretary

Personal information:

Name: Michael Y. Roleda
Contact Address: Marine Science Institute, University of the Philippines
Email: myroleda@up.edu.ph; michael.roleda@daad-alumni.de

Academic qualifications:

2005	Dr. rer. nat. (Biology), University of Bremen, Germany
1997	MSc (Marine Science), Marine Science Institute-University of the Philippines
1988	BSc (Biology), University of San Jose-Recoletos, Cebu, Philippines

Professional positions held:

2018–present	Professor, Marine Science Institute, University of the Philippines
2013–2018	Researcher, Norwegian Institute of Bioeconomy Research (NIBIO) [*previously Bioforsk Norwegian Institute for Agricultural and Environmental Research*]
2010–2013	Research Fellow/Lecturer, Botany, University of Otago, New Zealand
2009–2010	Algal Physiologist, Scottish Association for Marine Science, Scotland
2006–2009	Research Associate/Lecturer, Institute for Polar Ecology, Kiel, Germany
2005–2006	Postdoctoral Scientist, Alfred Wegener Institute, Bremerhaven, Germany
2001–2005	Researcher, Biologische Anstalt Helgoland, Helgoland, Germany
2000	Director, Marine Biological Station-DLSU, Batangas, Philippines
1997–2000	Associate Professor, De La Salle University (DLSU), Manila, Philippines
1992–1997	Research Associate, Marine Science Institute, University of the Philippines

Other positions:

- Associate Editor, Journal of Phycology (Wiley)
- Associate Editor, Marine Biology (Springer)
- Associate Editor, Frontiers in Marine Science
- Guest Editor, Frontiers in Marine Science; Frontiers in Microbiology

Research Interest:

Algal stress physiology; applied phycology; biodiversity; cultivation, selective breeding and hybridization; ecophysiology; ecotoxicology; photobiology; reproductive biology

Publication summary:

I have **91** ISI journal articles (33 first author, 2 single author, 15 senior/last author and 41 co-author). My H-index is 33 (source: WoS, https://apps.webofknowledge.com accessed 20.02.2022). I am a co-author of 6 book chapters and first author on one book chapter.

Other publications include eight peer-reviewed articles in non-ISI-journal papers, conference proceedings and scientific reports, and 4 scientific reports and extended abstracts. Participation in international and national conferences consist of 45 and 23 oral presentations, respectively, and 32 posters. I have been invited to present 20 public lectures, seminars, and workshop.

Teaching experience:

2018–present	University of the Philippines, Philippines
2010–2012	University of Otago, New Zealand
2006–2009	Christian Albrecht Universität zu Kiel, Germany
1997–2000	De La Salle University, Philippines

Graduate student supervision:

1. Bienson Ceasar V. Narvarte (MS Marine Science, 2020) Inorganic carbon utilization of tropical calcifying macroalgae and the survival of rhodolith *Sporolithon* sp. (Sporolithales: Rhodophyta) under long-term low pH conditions. Marine Science Institute, University of the Philippines.
2. Pablo Leal (PhD Botany, 2016) Effects of ocean acidification, temperature and copper on the development of early life stages of the native kelp *Macrocystis pyrifera* and the invasive kelp *Undaria pinnatifida* from southern New Zealand. Department of Botany, University of Otago, New Zealand. https://ourarchive.otago.ac.nz/handle/10523/6791
3. Pamela Fernandez (PhD Botany, 2015) The effects of ocean acidification on photosynthesis, growth and carbon and nitrogen metabolism of *Macrocystis pyrifera*. Department of Botany, University of Otago, New Zealand. https://ourarchive.otago.ac.nz/handle/10523/6227

4. Yuanyuan Feng (PhD Botany, 2014) Environmental controls on the physiology of the marine coccolithophore *Emiliania huxleyi*. Department of Botany, University of Otago, New Zealand. https://ourarchive.otago.ac.nz/handle/10523/5589

Current Research projects:

2022–2023 Selection and curation of safe and healthy eucheumatoid seedlings for the future (SecureFuture). Safe Seaweed Coalition

2021–2022 Selection of new strains of *Kappaphycus alvarezii* from wild individuals and their progeny. Sea6 Energy

2018-2022 Environmental controls on the physiology of different strains of Philippines eucheumatoids. *UP-MSI inhouse project.*

2020-2021 Growth performance of newly established *Kappaphycus alvarezii* seedstock from wild population. *Outright Research Grant, UP-Diliman.*

2020–2021 Selection and distribution of new strains of Kappaphycus alvarezii developed from spores of local wild populations to restore seaweed farming productivity in the Philippines GSS/RF/047 GCRF Global Seaweed Star, UK Research and Innovation

2020-2022 Nutrient physiology of commercially important eucheumatoids: two-phase cultivation strategy to optimize biomass and carrageenan yield and quality. *Balik PhD, UP-System.*

2019-2021 Diversity of wild and farmed eucheumatoids in the Philippines at the intra- and inter-specific scale: population genetics of *K. alvarezii* and molecular taxonomy of eucheumatoids. *UK Research and Innovation, Global Challenges Research Fund.*

2019-2021 Diversity of and population genetics of wild *Kappaphycus alvarezii* in Guiuan, Samar, Philippines. *ECWRG, UP-System*

2019-2020 Predicting Responses between Ocean Transport and Ecological Connectivity of Threatened ecosystems in the WPS (PROTECT WPS): Seaweed Component. *DENR-BMB.*

2018-2021 Coastal Acidification: How it affects the Marine Environment and Resources in the Philippines. *DOST.*
Project 2: Impacts of ocean acidification on the base of the marine food web and their effects on marine production
Project 3: Possible influence of ocean acidification on specific reef resources

Service to the scientific community:

i. Reviewing journals:
2008-2020: 20-50 annual invitations to review manuscripts for top journals such as *Oikos*; *Global Change Biology*; *Limnology and Oceanography*; *Planta*; *Plant Cell and Environment*; *Marine Ecology-Progress Series*; *Oecologia*; *Marine Biology*; *New Phytologist*; *Marine Biotechnology*; *PloS One; Proceedings of the Royal Society B; Journal of Experimental Botany*; *Photochemistry and Photobiology*; *Scientific Reports*, among others.

ii. Reviewing international research grants:
Fonds de la Recherche Scientifique, Belgium; Estonian Research Council, Estonia; COST Action, European Cooperation in Science and Technology, EU; National Science Foundation, U.S.A. (NSF); Commission for Scientific and Technological Development, Chile (CONICYT); Deutsche Forschungsgemeinschaft, Germany (DFG); National Science Centre (Narodowe Centrum Nauki), Poland; Alaska Coastal Marine Institute, U.S.A. (CMI); Czech Science Foundation, Czech Republic (GAČR); The Netherlands Organization for Scientific Research (NOW)

PUBLICATION LIST (Top 10 cited publications)

1. Bartsch I, Wiencke C, Bischof K, Buchholz CM, Buck BH, Eggert A, Feuerpfeil P, Hanelt D, Jacobson S, Karez R, Karsten U, Molis M, **Roleda MY**, Schubert H, Schumann R, Valentin K, Weinberger F, Wiese J (2008) The genus *Laminaria* sensu lato: recent insights and developments. European Journal of Phycology 43: 1-86.
2. **Roleda MY**, Slocombe SP, Leakey RJG, Day JG, Bell EM, Stanley MS (2013) Effects of temperature and nutrient regimes on biomass and lipid production by six oleaginous microalgae in batch culture employing a two-phase cultivation strategy. Bioresource Technology 129: 439-449.
3. Boyd PW, Dillingham PW, McGraw CM, Armstrong EA, Cornwall CE, Feng Y, Hurd CL, Ringold-Gault M, **Roleda MY**, Timmins-Schiffman E, Nunn BL (2016) Physiological responses of a Southern Ocean diatom to complex future ocean conditions. Nature Climate Change 6: 207-213.

4. **Roleda MY**, Morris JN, McGraw CM, Hurd CL (2012) Ocean acidification and seaweed reproduction: increased CO_2 ameliorates the negative effect of lowered pH on meiospore germination in the giant kelp *Macrocystis pyrifera* (Laminariales, Phaeophyceae). Global Change Biology 18: 854-864.
5. Gómez I, Wulff A, **Roleda MY**, Huovinen P, Karsten U, Quartino ML, Dunton K, Wiencke C (2009) Light and temperature demands of benthic marine micro- and macroalgae in polar regions. Botanica Marina 52:593-608
6. **Roleda MY**, Boyd PW, Hurd CL (2012) Before ocean acidification: calcifier chemistry lessons. Journal of Phycology, 48: 840-843.
7. Holtzinger A, **Roleda MY**, Lütz C (2009) The vegetative arctic freshwater green alga *Zygnema* is insensitive to experimental UV exposure. Micron 40:831-838.
8. Wiencke C, **Roleda MY**, Gruber A, Clayton MN, Bischof K (2006) Susceptibility of zoospores to UV radiation determines upper depth distribution limit of Arctic kelps: evidence through field experiments. Journal of Ecology 94: 455-463.
9. Duarte B, Martins I, Rosa R, Matos AR, **Roleda MY**, Reusch TBH, Engelen AH, Serrão EA, Pearson GA, Marques JC, Caçador I, Duarte CM, Jueterbock A (2018) Climate Change Impacts on Seagrass Meadows and Macroalgal Forests: An Integrative Perspective on Acclimation and Adaptation Potential. Frontiers in Marine Science 5:190.
10. Tatters AO, **Roleda MY**, Schnetzer A, Fu F, Hurd CL, Boyd PW, Caron DA, Lie AAY, Hoffmann LJ, Hutchins DA (2013) Short- and long-term conditioning of a temperate marine diatom community to acidification and warming. Philosophical Transactions of the Royal Society B: Biological Sciences 368: art. No. 20120437

Dr. Irene B. Rodriguez
The Marine Science Institute
University of the Philippines Diliman
Quezon City, Philippines
Email: ibrodriguez@msi.upd.edu.ph
ORCID ID: https://orcid.org/0000-0001-9848-9526
Lab website: https://msi-ibrlab.com/

Education

Dr. *rer. nat.* Chemistry	Karl-Franzens University Graz, Austria
MS Chemistry	University of the Philippines Diliman, Quezon City, Philippines
BS Chemistry	University of the Philippines Los Baños, Laguna, Philippines

Research and Teaching Experience

01/2019 – Present	**Associate Professor**, The Marine Science Institute, University of the Philippines Diliman (UP Diliman), Quezon City, Philippines
04/2012 – 12/2018	**Postdoctoral Research Fellow**, Research Center for Environmental Changes, Academia Sinica, Taipei, Taiwan
11/2008 – 5/2015	**Assistant Professor**, Institute of Chemistry, UP Diliman, Quezon City

Awarded Research Grants (2019 to present)

1. UK-RI: Selection and distribution of new strains of *Kappaphycus alvarezii* developed from spores of local wild populations to restore seaweed farming productivity in the Philippines (New Kappa Star); 2020-2021; Co-Project Leader; Budget: £ 49,995.17.
2. DOST-PCAARRD: Trace metals and nutrients from aerosol deposition: Impacts on community structure and occurrence of harmful algal blooms (MEDIATE-HABs); 2020-2023; Project Leader; Budget: PhP 4,783,784.00
3. DOST-PCAARRD: Macronutrient, carbon cycling, and trace metals: Impacts on phytoplankton community structure and toxin production of harmful algal blooms (Trace-HABs); 2020-2023; Project Leader; Budget: PhP 12,508,077.00
4. DOST-PCAARRD: Metallomics and Inorganic Biogeochemistry Facility for Research and Development; 2020-2022; Project Leader; Budget: PhP 19,850,000.00
5. UP-MSI: Investigating the role of nickel availability in the hydrogen production by the nitrogen-fixing cyanobacterium *Trichodesmium erythraeum*; 2020; Project Leader
6. UP-MSI: Influence of Fe availability and temperature variability on superoxide dismutase production of Symbiodiniaceae; 2019; Project Leader

Scholarly Development: Publications (representative)

1. Chen CC, Rodriguez IB, Chen YL, Zehr JP, Chen YR, Hsu SD, Yang SC, and Ho T-Y (2022). Nickel superoxide dismutase protects nitrogen fixation in *Trichodesmium*. Accepted paper in *Limnology & Oceanography Letters*.
2. Reich HG, Tu W-C, **Rodriguez IB**, Chou Y, LaJeunesse TC and Ho T-Y (2021). The importance of iron availability for coral symbionts' response to heat stress. *Journal of Phycology*, 57, 3-13, 10.1111/jpy.13078-20-150.
3. Reich HG, **Rodriguez IB**, LaJeunesse TC and Ho T-Y (2020). Endosymbiotic dinoflagellates pump iron: differences in iron and trace metal needs among the Symbiodiniaceae. *Coral Reefs*, https://doi.org/10.1007/s00338-020-01911-z.
4. Tuo S-H, **Rodriguez IB** and Ho T-Y (2020) H_2 accumulation and N_2 fixation reduction by Ni limitation in *Cyanothece*. *Limnology and Oceanography*, https://doi.org/10.1002/lno.11305
5. Li T, Lin X, Yu L, Lin S, **Rodriguez IB** and Ho T-Y (2019). RNA-seq profiling of *Fugacium kawagutii* reveals strong metabolic response and impacts on symbiosis potential of deficiencies of iron and other trace metals. *Science of the Total Environment*, https://doi.org/10.1016/j.scitotenv.2019.135767
6. **Rodriguez IB** and Ho T-Y (2018) Trace metal requirements and interactions in *Symbiodinium kawagutii*. *Frontiers in Microbiology*, 9:142. doi: 10.3389/fmicb.2018.00142
7. **Rodriguez IB** and Ho T-Y (2017) Interactive effects of spectral quality and trace metal availability on the growth of *Trichodesmium* and *Symbiodinium*. *Plos ONE*, 12:e0188777. doi: https://doi.org/10.1371/journal.pone.0188777



REY Y. CAPANGPANGAN, Ph.D.
PRC: 0003670

rey.capangpagan@msunaawan.edu.ph
09778203227
ORCID: 0000-0002-7078-9903
WoS Researcher ID: AAX-7641-2021
Citations: 132; h-index: 6

BIONOTE

Rey Y. Capangpangan is a Professor of Chemistry at MSU-Naawan, who specializes in the wide applications of nanotechnology and Material Science. His research interests are in areas of synthesis, characterization, and applications (biomedical and environmental) of various polymers and nanomaterials. Dr. Capangpangan is able to publish several scientific papers in ISI/Scopus indexed journals. He also received numerous awards, including the Presidential Award for Excellence in Research, Presidential Award for Excellence in Resource Generation, Outstanding Faculty, National Awardee (PCHRD-NHRS), among others. He also received a Research Fellowship in the Hebrew University of Jerusalem, Israel. He is actively involved in mentoring young researchers.

EDUCATION

2011 – 2015
Ph.D. in Chemistry
National Tsing Hua University
Taiwan

2005 – 2008
M.Sc. in Chemistry
University of the Philippines
Diliman

1997-2001
B.Sc. in Chemistry
MSU-Iligan Institute of Tech.

EXPERTISE

Sensor Fabrication
Nanoscience and Material Science
Polymer Chemistry
Environmental Monitoring
Green Synthesis
Proteomics
Nanomaterial Characterization

REY YONSON CAPANGPANGAN, Ph.D., RCh

Professor VI/ Chemist

PROFESSIONAL EXPERIENCE

Professor VI (Aug. 2021 – Present)
Mindanao State University – Naawan, Misamis Oriental, 9023, Philippines

Professor I (June 2004 – July 2021)
Caraga State University – Ampayon, Butuan City, 8600, Philippines

Laboratory Chemist (October 2001 – June 2004)
TECHNOLAB Laboratory Services – Cabancalan, Mandaue City, Cebu 6014, Philippines

RECENT JOURNAL PUBLICATIONS

Latest Journal Publications

1. Photocatalytic Activity of Cellulose Nanocrystals/Zinc Oxide Nanocomposite Against Thiazine Dye under UV and Visible Light Irradiation, *ASEAN Journal of Chemical Engineering* (JUST ACCEPTED MANUSCRIPT)
2. Single-Dip Colorimetric Detection of Cyanide using Paper-based Analytic Device based on Immobilized Silver Nanoparticles, *International Nano Letters* (JUST ACCEPTED MANUSCRIPT)
3. Variability and Source Characterization of Regional PM of Two Urban Areas Dominated by Biomass Burning and Anthropogenic Emission, *Aerosol and Air Quality Research*, 2022, **22**(6), 220026 (DOI: 10.4209/aaqr.220026)
4. Assessment of Impact of Meteorology and Precursor in Long-Term Trends of PM and Ozone in a Tropical City, *Aerosol and Air Quality Research*, 2022, **22**(1), 210269
5. Preparation of spin-coated poly(vinyl alcohol)/chitosan/gold nanoparticles composite and its potential application for colorimetric detection of cyanide in water, *Polish Journal of Environmental Studies*, 2022, **31**(2), 1-8 (DOI: 10.15244/pjoes/140276)
6. Improvised centrifugal spinning for the production of polystyrene microfibers from waste expanded polystyrene foam and its potential application for oil adsorption, *Journal of Engineering and Applied Science* (Springer), 2021, **68**, 25
7. Functionalized silver nanoparticle-decorated paper sensor for rapid colorimetric detection of copper ions in water, *Funct. Compos. Struct.*, 2021, **3**, 035007 (DOI: 10.1088/2631-6331/ac25e9)

RESEARCH EXPERIENCE

Projects handled/implemented

1. Project Staff, L3 – DOST-GIA CRADLE Program (2021 – 2022)
2. Project Leader – DOST-PCIEERD Project No. 08905 (2020-2021)
3. Project Leader – DOST-PCIEERD Project No. 09021 (2020-2021)
4. Project Staff (Collaborator) – eAsia Project with MSU-IIT (2020-2022)
5. Program Leader – CHED Funded Program in Response to COVID-19 (2020)
6. Project Leader – CHED Funded Project under the DARE TO K12 Program (2018-2020)
7. Project Staff (Collaborator) – CHED Funded under the DARE TO K12 (2018-2020)
8. Principal Investigator – Industry-supported Research Project (PHILSAGA) (2018-2019)
9. Principal Investigator – Industry-supported Research Project (SRMI) (2017-2018)
10. Project Staff – NGO-funded project (Haribon Foundation) (2018)

For more information: Please visit the website: https://rycapangpangan.wixsite.com/reycaps

Capsule Curriculum Vitae

CURRICULUM VITAE
MICHAEL A. CLORES

Educational Background:

Doctoral degree — 2011 – 2015
De La Salle University – Manila
PhD in Biology

2002 – 2005
De La Salle University – Manila
PhD in Science Education, Major in Biology

Master's degree — 2000 – 2002
Ateneo de Naga University
Master of Arts in Education, Major in Curriculum and Instruction

College degree — 1993 – 1998
Universidad de Santa Isabel
Bachelor of Science in Physical Therapy

Special Training:

J-1 Exchange Scholarship: Visiting Professor/Scholar
Focus: SPECIFY Databasing & Natural History Collections Management Work, August to September 2018
Herpetology Division, Kansas University Biodiversity Institute and Department of Ecology and Evolution, Natural History Museum, University of Kansas, USA.

Employment Background:

July 2021 to Present
Campus Dean/Associate Professor
Partido State University – Caramoan Campus

July 2020 to March 2021
Professor
Graduate School
Bicol State College of Applied Sciences and Technology

June 2000 to June 2020
Professor
Ateneo de Naga University

Other Positions Held:

September 2020 to March 2020
Director
Research, Extension, Production, and Enterprise Development,
Bicol State College of Applied Sciences and Technology

Recently published/ prepared/ presented papers:

1. Bernstein, J.M., Bautista, J., Diesmos, A.C., **Clores, M.A.** Cuesta, M.A., Sanguila, M.B. Brown, R.M. (2021) [In Press]. Multivariate Characterization of Phenotypic Variation from throughout the Geographic Ranges

of Philippine False Coral Snakes: Two Species or Four? ***Ichthyology & Herpetology*** 109, No. 4, 2021, 000–000. DOI: 10.1643/h2021035

2. Brown, R.M., Bautista, J.B., ***Clores, M.A.*** Cuesta, M.A. (2021) *DENDRELAPHIS MARENAE* (Gauke's Bronze-back Tree Snake). DIET. ***Herpetological Review***, 52(1), 153-154.
3. **Clores, M.A.**, Bautista, J.B., Fernandez, J.B., Cuesta, M.A., Brown, R.M. (2020). Diversity and distribution of amphibians and reptiles in the Caramoan Island Group (CIG), Maqueda Channel, Southern Luzon, Philippines. ***Journal of Asia-Pacific Biodiversity***. 14 (2020)1-14. https://doi.org/10.1016/j.japb.2020.11.005
4. Brown, R. M., Meneses, C.G., Wood, P.L.J., Fernandez, J.B., Cuesta, M.A., **Clores, M.A.**, Tracy, C., Buehler, M.D. Siler, C.D. (2020). Unexpected Discovery of another New Species of Philippine False Gecko (Gekkonidae; *Pseudogekko*) from the Bicol Peninsula of Luzon Island. ***Herpetologica*** 76(3), 315-329, (9 September 2020). https://doi.org/10.1655/Herpetologica-D-19-00029.1
5. **Clores, M.A.**, Conde, M.A.A., Perez, J.D. (2020). Biomass Leaching and Dynamics of Nutrients, Microbial Abundance and Activity during the Decomposition of Seagrass *Cymodocea rotundata* Necromass. ***Applied Environmental Research***, 42(1): 1-13. https://doi.org/10.35762/AER.2020.42.1.1
6. **Clores, M.**, & Cuesta, M. (2019). Trophic Models of Seagrass Ecosystems in Maqueda Channel, Caramoan Peninsula, Philippines. ***Applied Environmental Research***, 41(3), 14-31. https://doi.org/10.35762/AER.2019.41.3.2
7. Sarcia, S.M., Ang, M. P., **Clores, M.A.**, Oropesa L.J.B. (2016). Grazing Activity of Collector Urchin (Tripneustes gratilla, Linnaeus 1958) on the Leaf Stratum of Seagrasses in Sabitang-Laya Ilsand, Caramoan Peninsula, Philippines. ***Bicol Science Journal*** (3) 1, 8-20.

Note: cor complete list of papers published, please see https://scholar.google.nl/citations?user=X9RLwPYAAAAJ&hl=de or https://www.researchgate.net/profile/Michael-Clores-2

Awards received:

- First Recipient of the **Fr. Frank X. Lynch SJ. Award for Most Outstanding Researcher.** Ateneo de Naga University. February 20, 2020.
- Recipient: **Fr. Raul Bonoan, S.J, and Jose Quiros Professorial Research Chair Award.** Ateneo de Naga University
- Recipient of the **Best Paper Award**. 2nd International Congress on Action Research, Action Learning (ARAL2017). De La Salle University, Manila. April 1, 2017.
- Recipient of **Outstanding Research Paper Award**. First Place (Faculty Research Category). CHED Council of Deans for Graduate Education (CODGE V), March 2016

Organizations:

- Associate Member, National Research Council of the Philippines (Approved: April 2022)
- Biology Teachers Association, Inc. (BIOTA, Inc.)-National Chapter. Position: Member, Board of Directors (2021-2023)
- CHED Regional Quality Assessor Team (RQAT) Member
- Asian Association for Biology Education (AABE) – Regular Member
- Philippine Association of Marine Sciences (PAMS) – Regular Member
- Philippine Society for Microbiology, Inc. (PSM) ~Bicol Chapter. Position: President (2007-2008) – Ex-Officio

RICHARD V. DUMILAG, PhD

Associate Professor V
Fisheries Department, Sorsogon State University – Magallanes Campus
Magallanes, Sorsogon 4705 Philippines
richard.dumilag@sorsu.edu.ph +639361585522

DEGREES

PhD Biology, Institute of Biology, University of the Philippines, Diliman, 2017
MSc Marine Science, The Marine Science Institute, University of the Philippines, Diliman, 2012
BSc, Pamantasan ng Lungsod ng Maynila, Intramuros, Manila, 2003

EMPLOYMENT HISTORY

Mindanao State University – Tawi-Tawi, Professor VI, 2019-21
Far Eastern University, Assistant Professor, 2013-19
UP-MSI, University Research Assistant, 2010-16

GRANTS

Project Leader, DOST-PCAARRD, Molecular characterization, selection, and production of high quality eucheumatoid cultivars in the Bangsamoro Autonomous Region in Muslim Mindanao (BARMM) under a NICER Program: Establishment of Seaweed Research and Development Center (SeaRDeC) to Support the Seaweed Industry in BARRM, 2020-2021

Co-Project Leader, United Kingdom Research and Innovation–Global Challenges Research Fund (UKRI- GCRF) 'GlobalSeaweedSTAR' Programme (Grant No.GSS/RF/015). Diversity of wild and farmed eucheumatoids in the Philippines at the intra- and inter-specific scale: population genetics of *K. alvarezii* and molecular taxonomy of eucheumatoids, 2019-2020

Project Leader, Far Eastern University Office of the Research. Establishment of the Far Eastern University Herbarium (FEUH) Phase 2: Molecular Analysis, Project Title 3: Molecular-assisted alpha taxonomy of Philippine economically important seaweeds, 2018-2019

Project Leader, Far Eastern University Office of the Research. Establishment of the Far Eastern University Herbarium (FEUH) Phase 1: Specimen Collection, Project 2: Building strengths of the FEUH: Rhodophyte diversity of Sorsogon, 2017-2018

Consultant, National Fisheries Research Development Institute, Plant Division (NFRDI). Diversity of *Gracilaria* in the Philippines, 2017

Systematics of Philippine Bangiales. International Association of Plant Taxonomy (IAPT) Research Grants Program in Plant Systematics, Bratislava, Slovakia, Travel Grant, 2016

Accelerated Science and Technology Human Resource Development Program, Student Research Support Fund, 2016

Project Leader, Far Eastern University Office of the Research. DNA barcoding, genetic variation and biogeography in northern Philippine species of *Porphyra* (Bangiales, Rhodophyta), 2015-2016

PUBLICATIONS

Dumilag RV, Crisostomo BA, Aguinaldo ZA, Hinaloc LAR, Liao LM, Roa-Quiaoit HA, Dangan-Galon F, Zuccarello GC, Guillemin M-L, Brodie J, Cottier-Cook EJ, Roleda MY. The diversity of eucheumatoid seaweed cultivars in the Philippines. Reviews in Fisheries Science and Aquaculture. In Press.

Dumilag RV, Javier RF. 2022. Ethnobotany of medicinal seaweeds of Ilocos Norte, Philippines. Philippine Journal of Science 151(3): 1135–1156.

Gamus GCV, **Dumilag RV**, Yap SL. 2021. Use and Distribution of *Bangia* cf. *fuscopurpurea* (Bangiales, Rhodophyta) in the Northern Philippines. Philippine Journal of Science 151 (S1) 73–77.

Brakel J., Sibonga RC, **Dumilag, RV**, Montalescot V... Gachon CMM. 2021. Exploring, harnessing and conserving marine genetic resources towards a sustainable seaweed aquaculture. Plants People Planet 3(4): 337–349.

Dumilag RV, Samuyag AMP, Robles RJF. 2021. First report of *Chamaebotrys proliferus* (Rhodymeniaceae, Rhodophyta) from the Philippines. Philippine Journal of Science 150(1):263–266.

Roleda M.Y., Aguinaldo ZA, Crisostomo BA, Hinaloc LAR, Projimo VZ, **Dumilag RV**, Lluisma AO. 2021. Discovery of novel haplotypes from wild populations of *Kappaphycus* (Gigartinales, Rodohyta) in the Philippines. Algae 36(1): 1–12.

Dumilag RV, Lin S-M, Zuccarello GC, Kraft GT. 2020. The identity of *Eucheuma perplexum* (Solieriaceae, Gigartinales) and its distinction from *Euchuema serra* as exemplified by a proposed new epitype. Phycologia 59(6): 497–505.

Ferrer MSR, Gomez RN, **Dumilag RV.** 2020. COI-5P gene sequence variation in Philippine *Hydropuntia edulis* (Gracilariaceae, Rhodophyta). Philippine Journal of Science 149 (3-b) 1135–1140.

Dumilag RV et al. 2020. The Ulvophyceae (Chlorophyta) of eastern Sorsogon, Philippines, including *Halimeda magnicuneata* sp. nov. (Bryopsidales). Botanica Marina 63(5): 439-45.

Dumilag, R.V. 2019. Edible seaweeds sold in the local public markets in Tawi-Tawi, Philippines. Philippine Journal of Science 148 (4): 803–811.

Dumilag RV, Nelson W, Kraft GT. 2019. Validation and phylogenetic placement of the Placentophoraceae fam. nov. (Gigartinales, Rhodophyta). Phycologia 58 (3): 269–275.

Dumilag RV, Aguinaldo ZA, Alcoriza VAM, Balucanag MPSB, Dulalia ART, Sayasa AR. 2019. DNA barcodes of *Caulerpa* species (Caulerpaceae, Chlorophyta) from the northern Philippines. Philippine Journal of Science 148 (2): 343–353.

Ferrer MSR, Gomez RN, Nacido CB, Moises MT, **Dumilag RV.** 2019. Genetic diversity of Philippine *Gracilaria salicornia* (Gracilariaceae, Rhodophyta) based on mitochondrial COI-5P sequences. Biologia 74 (6): 599–607.

Dumilag RV. 2019. Detection of *Pythium porphyrae* infecting Philippine *Pyropia acanthophora* based on morphology and nuclear rRNA internal transcribed spacer sequences. Journal of General Plant Pathology 85(1):72–78.

Dumilag RV. 2018. *Betaphycus gelatinus* and B. *philippinensis* (Gigartinales, Rhodophyta) are conspecific. Phytotaxa 372 (1): 22–34.

Dumilag RV & Yap SL. 2018. *Pyropia lunae* sp. nov. and *Pyropia islae* sp. nov. (Bangiales, Rhodohpyta) from the Philippines. Botanica Marina 61(5): 467–480.

Dumilag RV. 2018. Unmasking a cryptic ethnotaxon: a case study on the identity of *Dermonema virens* (Nemaliales, Rhodophyta) in the Philippines. Webbia: Journal of Plant Taxonomy and Geography 73(1):39– 96.

Dumilag RV, Gallardo WGM, Garcia CPC, You Y, Chavez AKG & Agahan L. 2018. Phenotypic and mtDNA variation in Philippine *Kappaphycus cottonii* (Gigartinales, Rhodophyta). Mitochondrial DNA Part A 29(6):951–963.

Dumilag RV & Monotilla WD. 2018. Molecular systematics and biogeography of Philippine foliose Bangiales (Rhodophyta). Journal of Applied Phycology 30(1):173–186.

Dumilag RV, Aguinaldo ZAA, Mintu CB, Quinto MP, Ame EC, Andres AC, Monotilla WD, Yap SL, Cao EP, Vital PG, Fontanilla IKC. 2017. A review of the current taxonomic status of foliose Bangiales (Rhodophyta) in the Philippines. Phytotaxa 312(1):47–59.

Dumilag RV & Aguinaldo ZAA. 2017. Genetic differentiation and distribution of *Pyropia acanthophora* in the Philippines. European Journal of Phycology 52(1):104–115.

Dumilag RV, Aguinaldo ZAA, Mintu CB, Quinto MP, Ame EC, Andres AC, Monotilla WD & Yap SL. 2016. Morphological and molecular confirmation of the occurrence of *Pyropia tanegashimensis* (Bangiales, Rhodophyta) from Palaui Is., Sta. Ana, Cagayan, Philippines. Phytotaxa 255(1): 83 –90.

Dumilag RV, Salvador RC & Halling C. 2016. Genotype introduction affects population composition of native Philippine *Kappaphycus* (Gigartinales, Rhodophyta). Conservation Genetics Resources 8(4): 439- 441.

Dumilag RV, Orosco FL & Lluisma AO.2016. Genetic diversity of *Kappaphycus* species (Gigartinales, Rhodophyta) in the Philippines. Systematics and Biodiversity 14(5):441–451.

Dumilag RV & Lluisma AO. 2014. Resolving the phylogenetic affinities of *Kappaphycus inermis* within the genus *Kappaphycus* (Gigartinales, Solieriaceae) using mitochondrial and plastid markers. Phytotaxa 162(4):223 –231.

Dumilag RV, Liao LM & Lluisma AO. 2014. Phylogeny of *Betaphycus* (Gigartinales, Rhodophyta) as inferred from COI sequences and morphological observations on *B. philippinensis*. Journal of Applied Phycology 26(1):587 –595.

Dangan-Galon FD, **Dumilag RV** & Ganzon-Fortes E. 2013. Photosynthesis-irradiance curves of four marine macroalgae from Bolinao, Pangasinan and Calatagan, Batangas, Philippines. Scholarly Journal of Scientific Research and Essay 2(8): 134-138.

CURRICULUM VITAE

RONELIE CHATO SALVADOR
College of Agriculture, Fisheries and Natural Resources
UNIVERSITY OF EASTERN PHILIPPINES
Catarman 6400, Northern Samar

Phone : 09054852744
Email address : roneliesalvador@yahoo.com



Date and Place of Birth : Sept. 24, 1957, Surigao City
Citizenship : Filipino
Civil Status : Single
Sex : Female

Degrees :

Ph.D. Fisheries (Aquaculture) – UP in the Visayas, Miag-o, Iloilo, Nov. 2004
MSc. In Fisheries (Aquaculture) – UP in the Visayas, Miag-ao, Iloilo, 1993
BSc. Fisheries - Univ. of Eastern Philippines (UEP), Catarman, N. Samar , 1981

Work Experiences :

- Prof. 6 - UEP (Jan. 1, 2017 to Present)
- Prof. III – UEP (Sept. 1, 2014 to Dec. 31, 2016)
- Asso. Prof. V- (UEP) (Oct. 1, 2012 to Aug. 30, 2014)
- Assoc. Prof. III – UEP (July 2009 to Sept. 30, 2012)
- Postdoctoral researcher on seaweed biotechnology- Marine Science Institute, Univ of the Philippines, Diliman, Quezon City (June 2010 to Jan. 2011)
- Assoc. Prof. II - UEP (Oct. 2005 to June 2009)
- Assoc. Prof. III - UEP (July 1995 to Sept 2005)
- Visiting Aquaculture Research Specialist - SEAFDEC-AQD, Tigbauan, Iloilo (Oct. 2005 to Dec. 2007)
- Asst. Prof. II - UEP (Jan. 1995 to June 1995)
- Instructor III - UEP (Mar. 1993 to Dec. 1994)
- Instructor II - UEP (Aug. 1991 to Feb. 1993)
- Instructor I - UEP (Nov. 1988 to July 1991)
- Technical Research Asst. - BFAR-NSIRDP (July 1988 to Sept.1988)
- Instructor - UEP (June 1986 toJune 1987)
- Sr. Fishery Biologist - BFAR, Northern Samar (Nov. 1984 to June 1986)
- Project Officer - Ministry of Human Settlements (Feb. 1983 to Feb.1984)
- Technical Staff Assistant - Ministry of Human Settlements (April 1982 to Feb.1983)

Present designations :

- Dean, College of Agriculture, Fisheries and Natural Resources, UEP
- Coordinator for Agronot studies, UEP

Current research involvement:

Program Leader – Saganang Buhay sa Ilog (SALOG) research program (2020-2022)
Funding agency: National Research Council of the Philippines
Study Leader - Production and profitability of Parasol method of *Kappaphycus* farming

Major Researches completed:

- Development, validation and adoption of culture-based fisheries in Catubig and Pambujan Rivers, Northern Samar, Philippines
- Assessment of culture performance of *Kappaphycus* spp. in Northern Samar
- Histochemical profiling of 'ice-ice' disease infected *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta)
- Development of aquatic plant-based diets for *Kuhlia rupestris*
- Assessment of culture performance of indigenous freshwater clam, *Batissa* sp. in Catubig River
- Stock assessment and culture of freshwater prawn, *Macrobrachium rosenbergii*, in Northern Samar
- Inventory/assessment of seaweed resources of Northern Samar, Philippines
- Culture of tilapia hybrid in marine waters
- Growth, survival and bacterial load of sex-reversed Tilapia hybrid (*Oreochromis niloticus* x *Oreochromis mossambicus*) in seawater netcages
- Micropropagation of *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta)
- Testplanting of selected *Kappaphycus* strains in Northern Samar
- Generation of novel *Kappaphycus* varieties or strains from fused protoplasts

Research work experiences in other institutions:

June 2009-Jan.2010	-	Postdoctoral researcher on seaweed biotechnology, Marine Science Institute, Univ of the Philippines, Diliman, Quezon City
Oct. 2005- Dec. 2007	-	Visiting Aquaculture Research Specialist (SEAFDEC-AQD, Tigbauan, Iloilo); conducted research on strain improvement of farmed carrageenan-producing marine macroalgae

List of Technical Papers Published:

A. **Full Papers**

Tan KC, LB Mante, AD Mante and **RC Salvador** (2020). Yields from rice-fish farming at different water depths and fish stocking densities in wetland rice ecosystem. Pacific Journal of Science and Technology 4 (1): 28-34.

Dumilag, RV, RC **Salvador** and C. Halling (2016). Genotype introduction affects population composition of native Philippine *Kappaphycus* (Gigartinales, Rhodophyta). Conservation Genet Resour. DOI 10.1007/S12686-016-0591-2.

***Chato-Salvador* RC**, Tablizo FA and LLuisma AO (2014) Non-enzymatic isolation of somatic cells from *Kappaphycus* spp. and *Eucheuma denticulatum* (Solieriaceae, Rhodophyta). European Journal of Phycology, 49:4,486-492, DOI: 10.1080/09670262.2014.977963.

Salvador RC and Serrano AE (2014). Germination and growth of somatic cells of Philippine strains of *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta). ELBA Bioflux 6(1):36-45.

Vista, EB and **Salvador RC** (2014). Socio-economic status of artisanal fishermen in two rivers in Northern Samar , Philippines. Proceedings of the Second International in Fisheries Sciences. Colombo, Sri Lanka, July 30-31, 2014.

Salvador RC and Serrano AE (2005) Isolation of protoplasts from tissue fragments of Philippine cultivars of *Kappaphycus alvarezii* (Doty) Doty (Solieriaceae, Rhodophyta). J. Appl. Phyco. 17(1): 15-22.

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Dureza LA, Gempis CC, **Salvador RC,** Sombero RP and Belmonte BD (1994). Effects of selected factors on the growth and survival of Tilapia hybrids raised in cages in saline waters. In: LG Villacorta and LA Dureza (eds) Tilapia Farming: Genetic improvements and advances on culture technology. Proceedings of the Third National Symposium and workshop on Tilapia Farming. UPVisayas, Miag-ao, Iloilo.

Salvador RC, Gabona SF and Salvador AA (1994). Grouper species in the waters off the Northern Coast of Samar Island, Philippines. Pacific J. of Science and Technology 8: 29-33.

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Salvador RC and Salvador AA (1992). Culture trial of red Tilapia (*Oreochromis* sp) in polyethylene tanks. Pacific J. of Science and Technology 7 (1&2) : 40-46.

B. Published Abstracts

Salvador RC, De la Peña, MR and Santos FF (2007) Farmed carrageenan-producing seaweeds of the Philippines. Programs and Abstracts 39th Annual FIMFS Convention. Visayas State University, VISCA, Baybay City, Leyte.

Lim MM, Porlaje RS and Chato-Salvador RC (2006) Rheological properties of carragenophyte farmed in Eastern Visayas, Philippines. 38th Annual FIMFS convention, University of San Carlos, Cebu City.

Chato-Salvador RC, Santos FF, Ceniza PB and Consumo CA (2006) Growth performance and carrageenan properties of Philippines *Kappaphycus* and *Eucheuma* cultivars cultured in land-based facilities. 2006 Visayas Zonal Network Annual R & D Review and Planning Workshop.

Salvador RC and Salvador AA (2002). Inventory and assessment of seaweeds in Northern Samar. Proceedings of 30th FIMFS Pearl Anniversary convention. UEP, Catarman, Northern Samar.

Salvador RC, Salvador AA, Azamar MSL, Orane NL and Sorio MZ (1994). Growth response of *Macrobrachium rosenbergii* juveniles to various indigenous feedstuffs. Proceedings of the National Seminar-workshop on Fish Nutrition and Feeds. SEAFDEC AQD, Tigbauan, Iloilo.

Salvador RC, Atencio MD and Honrade MB (1994). Effect of feeding rates on growth and survival of Tilapia (*Oreochromis mossambicus*) cultured in netcages in seawater. Proceedings of the National Seminar-workshop on Fish Nutrition and Feeds. SEAFDEC AQD, Tigbauan, Iloilo.

C. **Published and copyrighted books and manuals**

Chato-Salvador, RC. 2016. Laboratory Manual in Fish Health Management. University of Eastern Philippines, Catarman, Northern Samar, Philippines. 58 p.

Chato-Salvador, RC. 2016. Laboratory Manual in Aquatic Botany. University of Eastern Philippines, Catarman, Northern Samar, Philippines. 51 p.

Chato-Salvador RC, MR De la Peña and FF Santos. 2017. Farmed Carrageenan-producing Seaweeds of the Philippines. SPPI, ICCO and CFC. 82 p.

Chato-Salvador RC. 2017. Farming of Carrageenan-producing Seaweeds (A Practical Guide). SPPI, ICCO and CFC. 27 p.

Chato-Salvador, RC. 2017. Simplified Methods for Assessment of Mangrove Ecosystems. SPPI and UEP. 23 p.

Intellectual Property Products with IPR

- Method of extracting crude enzymes from Golden Apple Snail (***Pomacea canaliculata)*** **– Utility Model # 2/2021/050056**
- Crude enzyme extracts from Golden Apple Snail (Pomaceae canaliculata) - Utility Model (submitted to IPOPhil and still in publication in IPOPHL Official Gazette)
- A method to dissociate tissues and isolate somatic cells from tissues of the red seaweeds *Kappaphycus* spp. and *Eucheuma* spp. (still in publication in the IPOPHL Official Gazette)

Awards Received :

A. Non-academic distinction/recognition

Title of Award	Area	Awarding Body/Agency	Level
Award of distinction	Public Service	UEP	Institutional
Loyalty Award			
Best Paper in Resource Assessment and Management (1995)	Research	FIMFS	National
Outstanding Applied Research Award (Bronze) (1992)	Research	VICARP	Regional
Outstanding Basic Research Award (Bronze) (2000)	Research	VICARP	Regional
Best Poster Paper Award (2000)	Research	VICARP	Regional
AFMA R & D Paper Award (2004)	Research	DA-BAR	National
2005 CHED Republica Award	Research	CHED	Zonal
Best poster paper award (Oct. 2007)	Research	PCAMRD	Zonal
Best poster paper award (Nov. 2007)	Research	FIMFS	National
Best Research Paper -1st Prize (Nov. 2010)	Research	DOST-PCAMRD	Zonal
Best Research Paper- 1st Prize (Jan. 2011)	Research	NARRDS/DOST-PCAMRD	National
Best Paper (Capture Fisheries Category) -1st Place (Oct. 2012)	Research	NRFDI	National
Best Paper (Aquaculture Category)-2nd place (2013)	Research	NFRDI	National
Best Paper (1st Place) (July 2015)	Research	VICARP-RRDEN	Regional

Best Paper Award (Research Category) Dec. 2015	Research	VICARP-RRDEN	Regional
2017 Dr. Elvira O. Tan Award (Outstanding Published Paper- Aquatic Sciences Category) (June 2017)	**Research**	**PCAARRD-DOST**	**National**
2017 CSC Dangal ng Bayan Awardee	**Public Service**	**Civil Service Commission**	**National**
Best Scientific Poster in Agricultural Sciences (July 2018)	**Research**	**National Academy of Science and Technology**	**National**

B. Academic Distinctions:

College Scholar Award - 1st semester AY 1999-2000, UP in the Visayas, Miag-ao, Iloilo

University Scholar Award - 2nd semester AY 1999-2000 and 1st semester AY 2000-2001, UP in the Visayas, Miag-ao, Iloilo.

Study/Scholarship Grants received :

MSU Full scholarship grant - AY 1973-1978
PCAMRD Thesis Grant - AY 1992-1993
PCAMRD Doctoral Study Grant - AY 1999- 2003
Netherlands Fellowship Program - May – June 2009

Membership in Organizations :
Member - Network of Women in Fisheries (Winfish)
Member - U.P.Visayas Graduate Students in Fisheries Association (UPV-GRASFA)
Member - Federation of Institutions of Marine and Freshwater Sciences (FIMFS)
Member - Philippine Fisheries Society
Member- Philippine Fisheries Association
Member- Society of Fisheries technologists in Eastern Visayas
Member- Netherlands Fellows Foundation of the Philippines, Inc.
Founding Member - Asian Society of Icthyologists
Member - Biodiversity Conservation Society of the Philippines
Associate Member - National Research Council of the Philippines